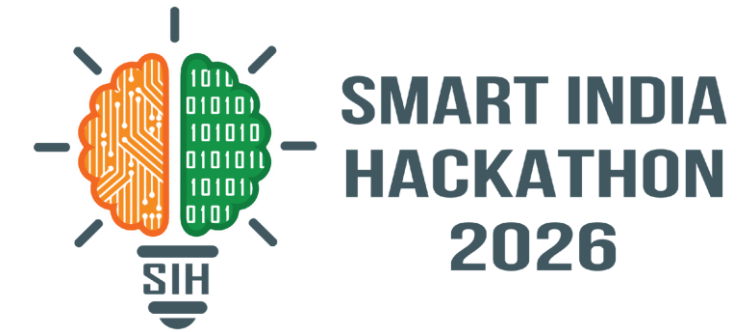
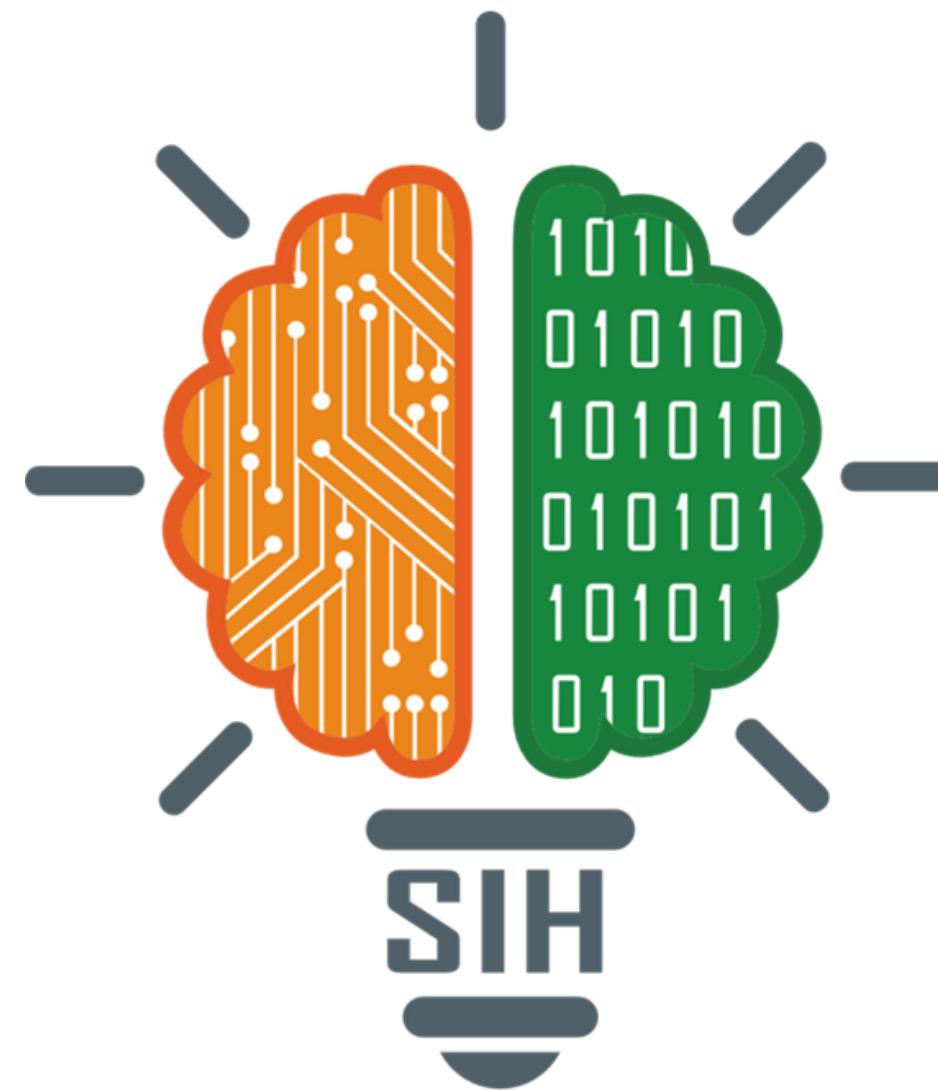


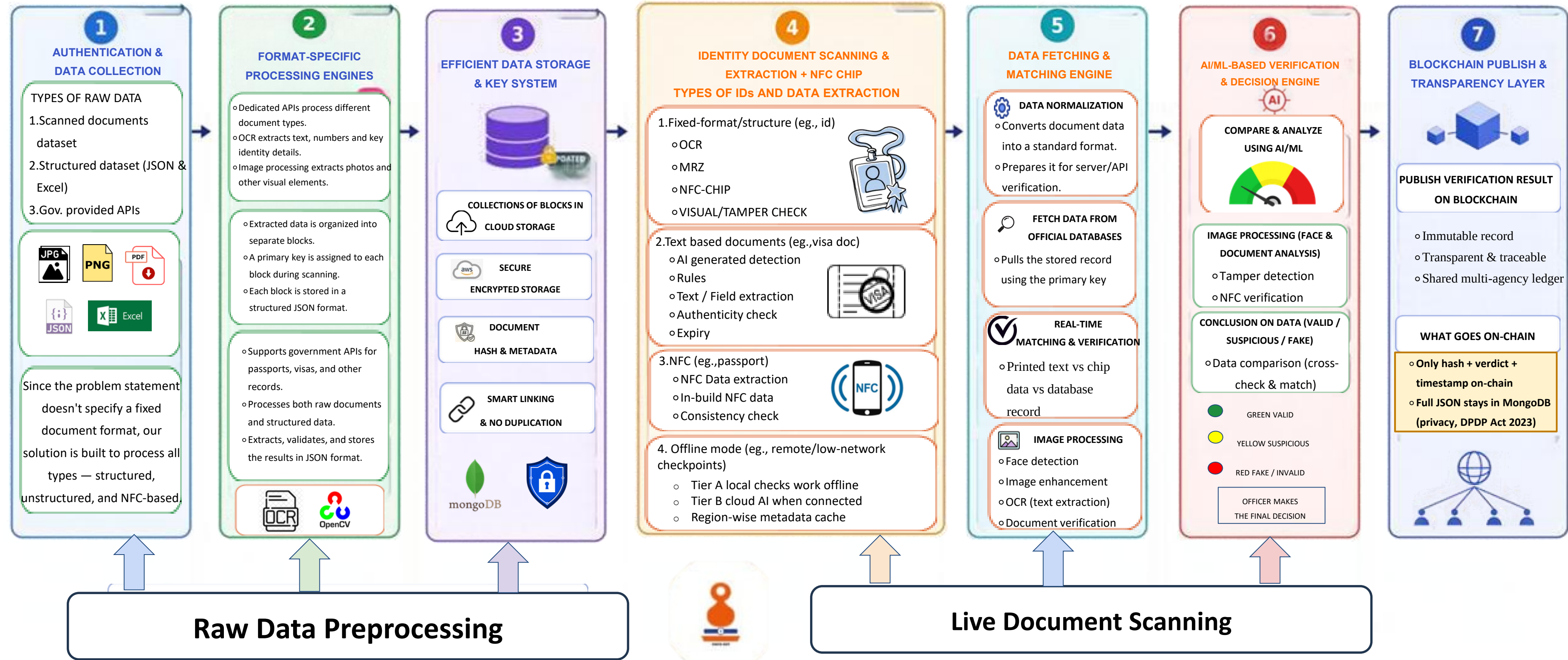
SMART INDIA HACKATHON 2026



VisionX

- Problem Statement ID – **SIH26188**
- Problem Statement Title- **AI-Based Fake Identity & Document Screening System**
- Theme- **Blockchain & Cybersecurity**
- PS Category- **Software**
- Team ID- **146366**
- Team Name – **SixBitss**





TECHNICAL APPROACH

PHASE 1 — DOCUMENT PROCESSING & DATA CAPTURE

1 DATA INGESTION & PREPROCESSING

- Tools:**
- OpenCV (cv2) — image manipulation
 - PyMuPDF (fitz) — PDF → image
 - Pillow, NumPy
 - UploadFile (JPG/PNG/PDF/WebP)
 - Batch folder ingestion
 - Direct JSON data ingest
 - API Attachments
 - Excel to JSON convertor
- Data Flow:** • Raw Image/PDF → Clean, Deskewed Image

2 DOCUMENT EXTRACTION APIs

- Tools:**
- PaddleOCR — accuracy fallback
 - OpenCV (cv2) — image manipulation
 - Regex (re) — strict field patterns
 - spaCy / NLTK — NER, dateutil
 - RapidOCR — fast edge engine
- Data Flow:** • Clean Image → Structured JSON (Name, ID No.)

```
_id: "8800 6030 0101"
aadhaar_number: "8800 6030 0101"
card_image_path: "storage/cards/8800_6030_0101.jpg"
dob: "01-09-1995"
face_photo_base64: "data:image/jpeg;base64,/9j/4AAQ
face_photo_path: "storage/faces/8800_6030_0101.jpg"
gender: "Female"
metadata: Object (3)
name: "Purvikapujari"
processed_at: "2026-09-08T13:28:01.037033"
source_file: "100front_blurred.jpg"
source_type: "demo_sample"
qr_data: null
```

Extracted Data in JSON Format

3 CLOUD DATABASE & STORAGE

- Tools:**
- MongoDB Atlas — cloud NoSQL DB
 - AWS S3 — raw image storage
 - motor (async), pymongo / boto3
 - Database: sih_identity_db
 - Collection: verified_documents
 - Upsert (prevents duplicate records)
 - Indexes: ID / name / processed_at
- Region-wise metadata cache** — offline backup + local download
- Data Flow:** • B-Tree Indexing (Primary Key) - Compound Indexing
- JSON Schema Validation

PHASE 2 — LIVE VERIFICATION & AI DECISION (at checkpoint)

4 LIVE ID PHOTO CAPTURE & SCANNING

- Tools:**
- OpenCV (VideoCapture) — live camera frames
 - MediaPipe — document detection & tracking
 - InsightFace + ONNX Runtime — face detection/embedding
 - OCR — text extraction (ID details)
 - High-resolution document image
 - Blur, glare & quality checks
 - Live document camera
 - Face alignment & detection
 - RapidOCR text extraction

5 TIER A — LOCAL CHECKS & FORENSICS

- Tools:**
- Works offline (no signal needed)
 - InsightFace (ArcFace) 512-D 1:1 + cosine similarity
 - Rapidfuzz text match (OCR/NFC vs cached record)
 - MRZ checksum + Verhoeff validation
 - NFC chip data vs printed data check
 - OpenCV/CNN forensics: font/kerning, emblem/seal placement, border geometry, ELA
 - Local baseline verdict: Valid / Suspicious / Needs connectivity
 - Query cached record by primary key

6 TIER B — LLM REASONING LAYER

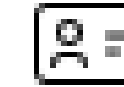
- Tools:**
- Requires connectivity (cloud enhancement)
 - Groq Cloud API — fast LLM inference
 - LangChain — prompt orchestration
 - Input: forensic flags + OCR/NFC/face scores
 - Anomaly tags & weighted risk fusion
 - Dynamic weight renormalization (if sensors unavailable)
 - Plain-English explanation for the officer
 - Government database cross-check
 - Human officer confirms final action

7 BLOCKCHAIN VERIFICATION & AUDIT

- Tools:**
- SHA-256 Merkle / hash chain
 - Timestamp + officer ID
 - Non-blocking queue
 - Immutable audit trail
 - Local append: target <10 ms
 - Exportable investigation / case record
 - Only verification proof/hash (not raw data)

PHASE 2 : TWO LOGINS ON APPLICATION

Border Checker Login



Fixed ID Scan



Document Scan

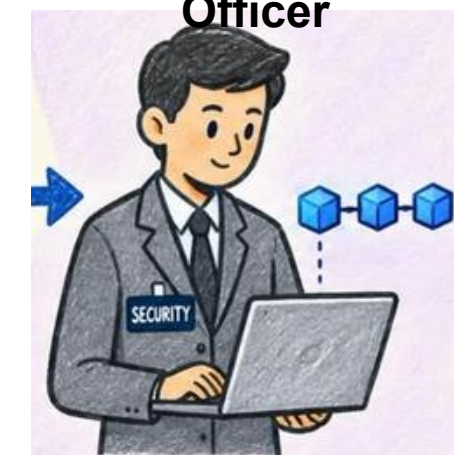


All Data Processed



NFC Chip Read

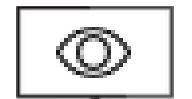
Security Officer



Phase 6 Risk Result



Added to Blockchain



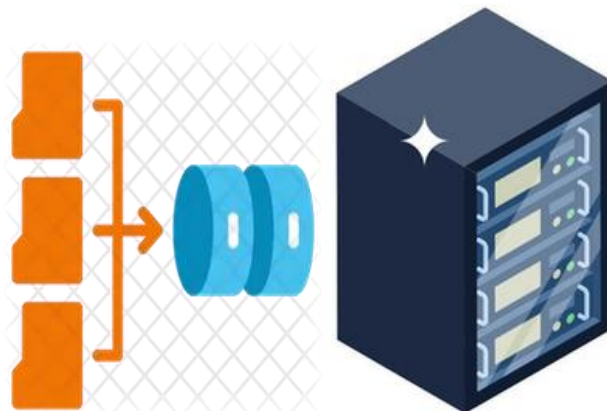
Views Chain Data

PHASE 1 - RAW DATA PROCESSING

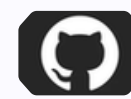
SCANNED DOCUMENTS

STRUCTURED DATA

GOV. APP'S



DATA SERVER



GITHUB REPO

PHASE 1 LINK-

https://github.com/vk9199kadam-hue/sih_phase1

PHASE 2 LINK -

https://github.com/vk9199kadam-hue/phase_2.git



You Tube Video QR





TECHNICAL + MAINTENANCE

FEASIBILITY

- FastAPI + OpenCV + RapidOCR + ArcFace
- OCR, NFC, face crop & secure storage
- MRZ, Verhoeff & QR validation
- Target: <2 sec processing

VIABILITY

- Adapter-based new document formats
- Reusable OCR & risk engines
- Standards can evolve without rebuilding



SECURITY + LEGAL

FEASIBILITY

- Confidentiality, Integrity & Availability
- Data minimization + masked storage
- Encryption + Role-Based Access Control
- Raw identity data kept off-chain

VIABILITY

- SHA-256 hash-chain for audit trail
- Supports electronic-evidence requirements
- Legal use follows applicable procedures



COMMERCIAL + FINANCIAL

FEASIBILITY

- Commodity servers + open-source stack
- Cloud AI optional; local checks need none
- Low checkpoint infrastructure cost

VIABILITY

- Reusable across multiple checkpoints
- Vendor-agnostic adapters
- Lower recurring licensing cost
- Scalable deployment



OPERATIONAL + WORKFORCE

FEASIBILITY

- Border-Detector app + Officer dashboard
- IDs, visas/permits + NFC verification
- One hosted server; new checkpoint = new queue
- Edge-first: local Tier A checks run offline
- RBAC for guard / officer / auditor

VIABILITY

- One hosted server; new checkpoint = new queue
- Decision-support, not automated gate
- Officer remains final decision-maker
- Explainable risk reasons



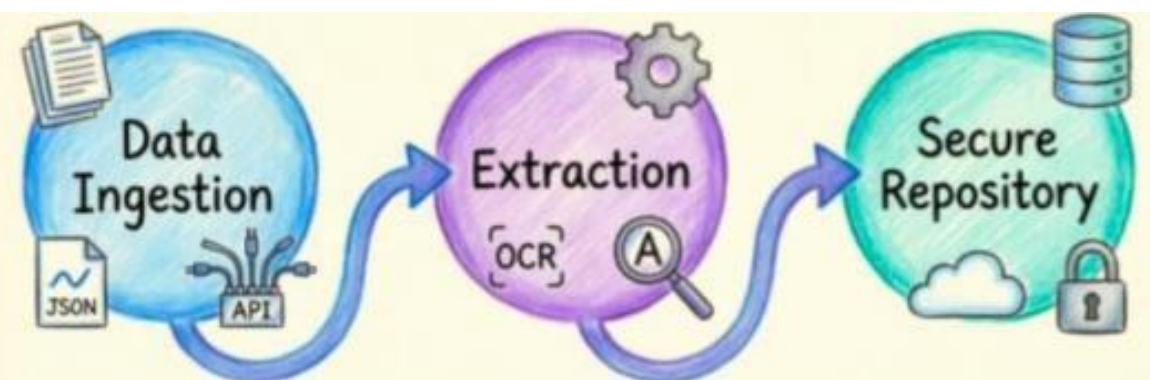
RISKS & MITIGATION

FEASIBILITY

- No real documents → synthetic + labelled data
- No live govt APIs → mock-to-real adapter
- Physical tamper blind spot → halo detector + referral
- Forwarded-image false alarms → provenance thresholds

VIABILITY

- Modular adapters reduce future integration risk
- Human review handles uncertain cases
- Audit trail supports traceability



DESIGNED FOR REAL DEPLOYMENT

Secure • Modular •
Explainable
Scalable across
checkpoints

Prototype note: built on synthetic data (no real documents) • government APIs simulated via mock adapter until access is granted



IMPACT & BENEFITS



FASTER VERIFICATION

- Reduced manual document checking
- Automated OCR, database lookup and validation
- From minutes to seconds (target: under 2 sec per traveller)

KPI: Verification time

STRONGER FRAUD DETECTION

- Combines document, face, NFC and AI signals
- Detects inconsistencies and anomalies
- Designed to flag forgeries (altered photos, tampered stamps)
- Tamper test: 48 of 50 altered documents flagged (96.0%), 1 of 50 clean documents falsely flagged (2.0%)

KPI: OCR field accuracy 98.2%
Face-match TPR 99.1% / FPR 0.9% (n = 100 synthetic docs)

BETTER OFFICER DECISION SUPPORT

- Green / Yellow / Red risk result
- Evidence-based verification information
- Human officer confirms final action

KPI: Review time

STANDARDIZED CHECKPOINTS

- Same verification workflow across all checkpoints
- Consistent rules and validation process
- Uniform decision-making nationwide

KPI: Exception / failure rate

TRACEABLE & ACCOUNTABLE SYSTEM

- Verification proof recorded with timestamp and officer ID
- Security officers can review verification checkpoints
- Immutable, blockchain-anchored log (anti-corruption)

KPI: Audit completeness

SCALABLE GOVERNMENT OPERATIONS

- Queue-based request processing
- Supports increasing verification workload
- Can scale from one server to multiple servers
- Handles high passenger volume

KPI: Requests/min + queue time

BETTER TRAVELLER EXPERIENCE

- Faster routine verification
- Less repetitive document handling
- Smoother checkpoint processing
- Reduced waiting time at borders

KPI: Median processing time

KEY OUTCOMES

- From minutes to seconds (operational speed)
- Data-driven risk assessment (accuracy & AI)
- Standardized screening nationwide (governance)
- Tamper-proof digital trail (blockchain)
- Secure & privacy-preserving verification

Feature	Current Manual / Legacy Systems	SixBitss
Real-Time Risk Triage (Green/Yellow/Red)	X	✓
Biometric / Chip Identity Check	X	✓
AI Risk Engine & Anomaly Detection	X	✓
Automated OCR / NFC Scanning	X	✓
Cross-Agency API Interoperability	X	✓
Immutable Blockchain Audit Trail	X	✓
Investigation and intelligence trail	X	✓



RESEARCH & REFERENCES



VISIONX — BORDER IDENTITY SCREENING
MINISTRY OF HOME AFFAIRS • SASHASTRA SEEMA BAL (SSB) • SIH26188

Install App / APK

ONLINE / CLOUD LINKED

SSB-IN-NPL-SECTOR-04

BORDER GUARD / CHECKPOINT TERMINAL

SECURITY OFFICER COMMAND & BLOCKCHAIN EXPLORER

1. Fixed Structured ID (Aadhaar) 1
2. Unstructured Data (Visa / Permit) 2
3. NFC Digital Chip (e-Passport) 3
4. Offline Regional Search (No Net) 4

1-CLICK VERIFICATION TEST PRESETS (REAL VS. FAKE):

Aadhaar Card 1 (Atharv) (0011 0022 0033)

Aadhaar Card 2 (Dhruva) (9876 5432 1002)

Aadhaar Card 3 (Haroon) (1234 1234 5555)

Aadhaar Card 4 (John Loyal) (1100 2200 3300)

Document & Live Camera Ingestion

Upload Image

Click to Upload Document Scan
Supports JPG, PNG, WebP or live camera capture

EXECUTE MULTI-SIGNAL INSPECTION

Intelligent Decision Support System

AYER Risk Engine

GREEN

Risk Score: 31.28 / 100

Recommended Clearance (E-Gate Eligible; Officer Confirms)

OFFICER ACTION

ACTIVE REASON CODES:
INVALID_VERHOEFF_CHECKSUM

Extracted Name	UNRESOLVED_SUBJECT
Aadhaar UID	UNRESOLVED_UID
Verhoeff Checksum	X CHECKSUM FAILED
DOB / Gender	01-01-1995 Male
ArcFace 512-D Biometrics	94.5% (Age Compensated)
Photo Halo Gradient	7.22 (OK)
Layout Forensics	Layout & Typography Validated
Blockchain Anchor	#1 (6acF0b0070ab6aFb...)

VISIONX — BORDER IDENTITY SCREENING
MINISTRY OF HOME AFFAIRS • SASHASTRA SEEMA BAL (SSB) • SIH26188

Install App / APK

ONLINE / CLOUD LINKED

SSB-IN-NPL-SECTOR-04

BORDER GUARD / CHECKPOINT TERMINAL

SECURITY OFFICER COMMAND & BLOCKCHAIN EXPLORER

Immutable Blockchain Merkle Ledger (DPDP Act 2023 Compliant)

CHAIN VERIFIED IMMUTABLE (4 Blocks Validated)

Refresh

Privacy Guarantee: In compliance with the Digital Personal Data Protection (DPDP) Act 2023, personal identity details remain strictly off-chain in encrypted storage. Only cryptographic document hashes, verification verdicts, and officer IDs are anchored to the blockchain.

Block #	Verdict	Risk Score	Modality	Officer ID	SHA-256 Merkle Block Hash	Timestamp
#3	RED	90	UNSTRUCTURED_VISA	GUARD_0482_SINGH	144329ad5daf1b55c5...	2026-10-02 11:12:37
#2	GREEN	10	UNSTRUCTURED_VISA	GUARD_0482_SINGH	0cda38c1e6954b06a9...	2026-10-02 11:12:24
#1	GREEN	31.28	FIXED_ID_AADHAAR	GUARD_0482_SINGH	30baec77b93d462bf3...	2026-10-02 11:12:18
#0	SYSTEM_INIT	0	GENESIS	ROOT_AUTHORITY	d1b1cb5871e1b12469...	2026-10-02 11:11:45



Prototype

Website + App



PrivIdEx Privacy-preserving and secure exchange of digital identity assets.

research.ibm.com — PrivIdEx

Localized Forgery Detection

Research on localized manipulation of facial and textual fields in digital identity documents. arxiv.org/abs/2605.09089

Privacy — DPDP Act 2023

Indian legal/privacy reference for handling digital personal data and data-protection controls. indiacode.nic.in — DPDP Act 2023

Identity Fraud Research

IDNet and related research support identity-document analysis and fraud-detection experimentation. arxiv.org/abs/2408.01690

Cybersecurity & Zero Trust

NIST CSF 2.0, NIST SP 800-207 and OWASP ASVS guide security architecture and verification. nist.gov/cyberframework

AI & Digital Forensics

Identity-document tamper-detection attacks, field-level forgery, and modern research arxiv.org/abs/2607.01442

Hyperledger Fabric

Reference architecture for permissioned blockchain networks and identity-based membership. research.ibm.com — Hyperledger Fabric

Government Digital Infrastructure

API Setu and DigiLocker provide reference points for government API and digital-document integration. apisetu.gov.in

Offline mode (no network): local verdict shown